CHEAT SHEET · DAY 3

# Survey analysis

Declaring the design and producing official estimates with their precision. Every command here was taught today; the deck shows each one at work on the LFS or the CPI.

## Why weights — Part 1

| | |
|---|---|
| `summarize wgt_final` | smallest, mean, largest weight |
| `display r(sum)` | the population the weights stand for |
| `tabstat w, statistics(min mean max) by(g)` | weights by group |
| `tabulate v [iweight = w]` | a quick weighted count, no SE |
| `mean v [pweight = w]` | weighted mean, SE still wrong |
| `svyset ... (Part 3)` | weights + design: the right SE |

## The design — Part 2

| | |
|---|---|
| `egen t = tag(a b)` | 1 on the first row of each a-b combination |
| `bysort g: egen n = total(t)` | count tagged rows per group |
| `tabulate n if t` | one row per group in the table |
| `assert abs(x - y) < 1e-4` | equal, allowing for rounding |
| `bysort g: assert x == x[1]` | x is constant within g |
| `v[1] v[_N]` | first and last row of the group |

## Svyset — Part 3

| | |
|---|---|
| `svyset psu [pweight = w], strata(s)` | declare the design |
| `svyset` | show the stored declaration |
| `svydescribe` | strata, PSUs, obs per PSU |
| `svydescribe v` | same, for rows where v is not missing |
| `... fpc(N)` | finite population correction (recognise) |
| `... brrweight() jkrweight()` | replicate weights (recognise) |

## Estimates — Part 4

| | |
|---|---|
| `generate f = (condition)` | a 0/1 flag, never missing |
| `svy: mean v` | design-based mean |
| `svy: total f` | number of people with f = 1 |
| `svy, subpop(wap): proportion v` | shares among the 15+ |
| `svy: ratio unemp/lf` | unemployment rate |
| `estat effects` | DEFF and DEFT of the last estimate |

## Subpopulations — Part 5

| | |
|---|---|
| `svy, subpop(f): mean y` | estimate for the group f = 1 |
| `svy, subpop(if a == 1 & b == 2): ...` | group built inline |
| `..., over(region)` | one estimate per category |
| `..., over(sex agegrp)` | every combination |
| `svy: proportion f (subpop)` | interval that stays in 0-1 |
| `never: svy: mean y if group` | wrong SE for small groups |

## Survey tables — Part 6

| | |
|---|---|
| `svy: tabulate v` | weighted proportions |
| `..., count se ci format(%9.0fc)` | people, SE, interval |
| `..., percent ci format(%5.1f)` | percentages with interval |
| `svy: tabulate a b, row` | b within each a |
| `svy: tabulate a b, column` | a within each b |
| `Design-based F, P` | the test to read |

## Precision — Part 7

| | |
|---|---|
| `estat cv` | CV in percent, after svy: |
| `estat size` | interviews and population per estimate |
| `estat effects` | DEFF and DEFT |
| `never [aweight] for a release` | wrong SE |
| `svyset with strata()` | or the SE changes |
| `pweight = wgt_final` | never wgt_base |

## The estimates table — Part 8

| | |
|---|---|
| `matrix list r(table)` | see the last estimate's results |
| `matrix T = r(table)'` | copy, regions as rows |
| `matrix R = T[1..., 1..2], T[1..., 5..6]` | keep columns |
| `matrix P = 100 * P` | percent |
| `matrix rownames P = a b c` | names for the release |
| `putexcel A1 = matrix(T), names` | into the open sheet |

Help inside Stata: `help` *command*, and the PDF manual linked at the top of each help page.